



ANALYTICS & DECISION SUPPORT CASE STUDY

Healthcare Operations Benchmarking & Risk Monitoring

An anonymized healthcare operations analysis showing how structured data rules, validation, benchmarking, and monitoring can turn complex operational data into clearer decision support.

The challenge

Operational data from multiple communities and reporting periods needed consistent definitions, validation, and comparison rules before leaders could distinguish meaningful patterns from ordinary variation.

What DOMAS did

Standardized source data, encoded explicit business rules, created normalized benchmarks, compared communities and reporting periods, and translated multiple risk signals into priorities for operational review and follow-up.

How the analysis supported decisions

1 Standardize & validate	2 Benchmark	3 Compare	4 Prioritize
Apply consistent source-data rules before comparison.	Create normalized measures across communities.	Review changes across locations and reporting periods.	Flag multiple signals that warrant closer operational review.

What was measured

Weight monitoring

- Percent of residents with at least one defined weight-change event
- Weight-change events per standardized resident count
- Median days between weigh-ins
- Percent exceeding the expected monitoring interval

Fall-related monitoring

- Repeat-fall rate within a defined follow-up window
- Falls per standardized resident count
- Median days between falls

What the analysis produced

Decision-support outputs

- Community-level benchmark tables and comparisons
- Period-over-period changes in monitoring cadence and event frequency
- Multiple-signal flags identifying communities for closer review
- Hypotheses for follow-up, including missingness or under-detection, staffing or process changes, seasonality, and case mix

Selected signals observed

Longer weigh-in intervals in many communities, increases in weight-change event frequency, and meaningful variation in repeat-fall patterns.

Structured rules, validated data, and comparable benchmarks improve operational decision support.

What this demonstrates for clients

DOMAS can take messy operational data, make the rules and assumptions explicit, build trustworthy comparison logic, and translate the results into structured decision support. The same approach can support recurring performance review, risk monitoring, program oversight, quality improvement, and other operational reporting needs.

Explore the full case study: [View full case study PDF](#) • [View project repository](#)

Portfolio note: This is an anonymized analytics case study using aggregate results and sanitized materials. No raw resident-level data are included, and the analysis should not be interpreted as clinical guidance.